

Andrew Peterson

QUANTUM RESEARCHER · SECURITY ENGINEER · SOFTWARE ARCHITECT

California, United States of America

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Experience

Qualcomm Institute

La Jolla, California

QUANTUM SOFTWARE ENGINEERING LEAD (INTERN)

Jul. 2022 - Dec. 2024

- Architected and spearheaded a video communication platform utilizing end-to-end quantum encryption via photonic circuitry, enhancing data security protocols.
- Optimized Grover's algorithm using Qiskit to solve complex binary constraint satisfaction problems, demonstrating practical applications for quantum search.
- Developed and evaluated Quantum Machine Learning (QML) models, implementing classification algorithms via quantum neural networks to improve predictive accuracy.

Rochester Institute of Technology Esports

Remote (Rochester, New York)

FULL-STACK DEVOPS ENGINEER & ADVISOR

Sep. 2020 - Jul. 2022

- Engineered and deployed scalable web resources using the MERN stack (MongoDB, Express, React, Node.js), leveraging Docker for consistent environment orchestration.
- Orchestrated cloud infrastructure on DigitalOcean Droplets, implementing Nginx reverse proxies and Docker Compose to ensure 99.9% uptime for high-traffic services.
- Streamlined development lifecycles by building CI/CD pipelines and monitoring suites, reducing deployment friction and enhancing system performance.

Extracurricular

Quantum Computing at UC San Diego

La Jolla, CA

CO-FOUNDER & PRESIDENT (2023 - 2024) & TECHNICAL CURRICULUM LEAD

Feb. 2021 - Dec. 2024

- Co-founded and scaled the first undergraduate quantum computing organization at UC San Diego, establishing a community for 100+ aspiring researchers and engineers.
- Developed a comprehensive technical curriculum for workshops covering foundations in quantum computing and Qiskit.

San Diego Capture the Flag (SDCTF)

Remote

LEAD INFRASTRUCTURE ENGINEER & CHALLENGE DESIGNER

Dec. 2020 - May 2022

- Engineered the backend for San Diego's largest annual cybersecurity competition, supporting 500+ global participants in its inaugural year.
- Provisioned cloud-native infrastructure using Terraform (IaC), managing a Postgres database and containerized challenge environments.
- Designed and deployed a Discord.JS-integrated submission bot to automate real-time flag validation and leaderboard updates.

Skills

Languages Python (Expert), C/C++, Java, TypeScript/JavaScript, SQL, Bash, LaTeX, Matlab, Rust

Systems & DevOps Docker, Kubernetes, Terraform, Linux Kernel, Distributed Systems, CI/CD, Nginx, Git, AWS, DigitalOcean

Software Engineering Data Structures & Algorithms (DSA), System Design, Design Patterns, Agile/Scrum, Unit Testing

Data Science & AI PyTorch, Scikit-Learn, Pandas, Computer Vision, NLP, Ollama (LLMs), Statistical Modeling, Jupyter

Infrastructure PostgreSQL, MongoDB, Redis, RESTful/GraphQL API Design, Microservices, Linux Server Administration

Certifications

2025 **Azure Data Fundamentals**, Microsoft

2022 **IT Fundamentals+**, CompTIA

Education

University of California: San Diego

La Jolla, California

BACHELORS OF SCIENCE IN COMPUTER SCIENCE

Sep. 2020 - Dec. 2024